

SHARATH S RAO

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SUMMARY

GenAI Systems Engineer with 2+ years building production-grade LLM systems serving 24,000+ users. Specializes in multi-agent architecture (LangGraph), RAG pipelines, context engineering, and Evaluation-Driven Development. Proven track record shipping reliable AI systems using CI-integrated eval pipelines, LLM-as-Judge, and structured quality gates. Equally comfortable across the full stack — AI systems, backend engineering, and QA — in early-stage product environments. Open-source contributor to the LangChain ecosystem.

TECHNICAL SKILLS

LLM Orchestration: LangChain, LangGraph, LangSmith, Ollama

Multi-Agent Systems: Multi-Agent Architecture, Agentic Workflows, Graph-based Workflow Orchestration

Evaluation & Reliability: Evaluation-Driven Development (EDD), LLM-as-Judge, RAGAS, Promptfoo, DeepEval, CI-integrated Evals, Hallucination Detection, Guardrails, Red Teaming, Output Validation

RAG & Retrieval: Retrieval-Augmented Generation, Semantic Search, Hybrid Search, MMR, HNSW, Embeddings, Chunking Strategies

Vector Databases: PostgreSQL pgvector, Pinecone

Prompt & Context Engineering: Prompt Engineering, Context Engineering, Structured Outputs, Function Calling, Tool Calling, Schema-Constrained Prompting

Open-Source Models: DeepSeek, Llama, local model deployment via Ollama

Performance: Batching, Parallel Processing, Async Pipelines, Model Routing, Retry/Fallback, Low-Latency Inference

Full Stack & Infra: Python, NestJS, Next.js, REST APIs, PostgreSQL, Docker, System Design (HLD/LLD), UML, Multi-tenancy

EXPERIENCE

Technical Head — AI, Full Stack & QA

Jan 2025 – Present

TestMySkills.ai · Zyni Innovations

Bangalore, India

- Multi-disciplinary founding engineer across AI systems, full-stack development, and quality engineering on a B2C/B2B skill assessment platform with multi-tenant architecture supporting enterprise clients.
- AI Systems:** Architected multi-agent LLM systems using LangGraph with modular, graph-based workflow orchestration for intelligent, adaptive assessment delivery.
- Pioneered Evaluation-Driven Development (EDD):** built a CI-integrated eval pipeline where known failure cases are stored in a DB and automatically replayed as pre-commit checks — enabling confident feature shipping 95%+ of the time.
- Built RAG pipelines achieving ~90% accuracy** using advanced chunking, embedding tuning, and MMR retrieval; monitored and iterated using LangSmith observability.
- Optimized semantic search** with PostgreSQL pgvector using HNSW indexing, improving retrieval relevance and downstream LLM response quality.
- Full Stack:** Built and maintained B2C and B2B product features across Next.js frontend and NestJS backend; designed and implemented multi-tenant architecture.
- QA Engineering:** Established quality processes, test coverage strategies, and automated testing pipelines across the platform.
- Scaled platform to 24,000+ users with stable performance using async pipelines, model routing, and fault-tolerant retry/fallback mechanisms.
- Led an 8-engineer team — owning architecture (HLD + LLD), sprint execution, and cross-functional delivery.

AI Engineer

Jun 2024 – Jan 2025

Zysk Technologies

Bangalore, India

- Built the AI-based Skill Mastery Evaluation System** — automated LLM-based scoring and feedback system serving 24,000+ users across skill assessment workflows.
- Reduced evaluation latency 80%** (5 min → 1 min) through batching and parallel LLM execution across concurrent workloads.
- Implemented **LLM routing by model availability and cost** — maintained output quality while measurably reducing inference spend.

- Built **LLM-as-Judge pipelines** for hallucination detection, automated output verification, and quality assessment at scale.
- Engineered **fault-tolerant inference** with structured retry logic, fallback model chains, and guardrails.
- **PushsendAI (Zysk)**: Designed and built a personalized product recommendation engine using reinforcement learning and matrix factorization on user purchase profiles; engineered prompt-based email personalization using open-source models. Pilot commended by Veronica Beard (US fashion brand).

AI Engineer Intern

Mar 2023 – Jun 2024

Zysk Technologies

Bangalore, India

- Developed **prompt engineering frameworks** for scalable personalized generation across multiple product use cases.
- Built **structured-output pipelines** using schema-constrained prompting prior to native function-calling support in LLM APIs.
- Worked on hallucination mitigation strategies and LLM controllability and steering techniques.
- Designed and evaluated prompt optimization strategies to improve accuracy, consistency, and controllability in LLM outputs.

PROJECTS

Zyskie — AI-Powered HRMS Platform

2026

Zyni Innovations · Product Owner & Lead Engineer · Team of 8

- Built a full GenAI-powered HRMS covering intelligent hiring, workforce management, and internal talent discovery.
- Owned **end-to-end system architecture** (HLD + LLD + UML diagrams) and led technical delivery across the full stack.
- Implemented **PostgreSQL pgvector with HNSW and MMR** for embedding-based semantic search — enabling expert identification, mentor matching, and AI-driven team formation.
- Deployed **open-source models (DeepSeek) via Ollama** for cost-effective, privacy-preserving local inference.
- Built LLM-powered hiring workflows: JD generation, candidate evaluation, and automated scoring pipelines.
- Integrated RBAC, n8n workflow automation, and SaaS connectors (Zoho, Keka, BambooHR) on a PostgreSQL backend.

EDUCATION

B.Tech in Information Science & Engineering

2020 – 2024

JSS Academy of Technical Education

Bangalore, India

- CGPA: 8.8 / 10.0

PUBLICATIONS

Hardware Inventory Management System Using IoT — IEEE (9 citations)

Interactive Framework for Querying Data from Large PDFs — IEEE (7 citations)

Traffic Control Management System for Emergency Vehicles — Telematique Journal

OPEN SOURCE

LangChain (langchain-ai) — Active contributor to the LangChain open-source ecosystem; contributed to the HANAVector self-query retriever module through hands-on internals review.

SPEAKING & COMMUNITY

Keynote Speaker — IEEE Student Chapter Inauguration, Bangalore (150+ attendees); spoke on the intersection of GenAI and metacognition in learning and knowledge systems.

Prompt Engineering Educator — Conducted 5+ sessions at KSIT, JSS, SJCE Mysore, Presidency University for student and faculty development programs; 30–50 attendees per session.

Head of Advisory Committee — Basavanagudi Government School.

AWARDS & RECOGNITION

Emerging Leader Award (2025) — Recognized for technical leadership and platform impact.

Captain of Change Agents — Leader of Internal techno cultural team of Zysk Technologies.